I

(1)

For each Ising site, there are only two possible spin states, which have energies $-h$ for spin up and h for spin down, respectively.

The single-site partition function is

$$Z_1 = e^{\beta h} + e^{-\beta h} = 2\cosh(\beta h).$$

Since the N Ising spins are independent when $J = 0$, the partition function of the whole system is

$$\boxed{Z = Z_1^N = [2\cosh(\beta h)]^N}.$$

(2)

The Helmholtz free energy is

$$F = -k_B T \ln Z.$$

Substituting the result from (1), we obtain

$$\boxed{F = -Nk_B T \ln[2\cosh(\beta h)]}.$$

(3)

The internal energy is

$$U = -\frac{\partial}{\partial \beta} \ln Z.$$

Therefore,

$$\begin{aligned} U &= -N\frac{\partial}{\partial \beta} \ln[2\cosh(\beta h)] \\ &= -Nh\tanh(\beta h). \end{aligned}$$

Thus,

$$\boxed{U = -Nh\tanh(\beta h)}.$$

(4)

The total magnetization is

$$M = -\left(\frac{\partial F}{\partial h}\right)_T.$$

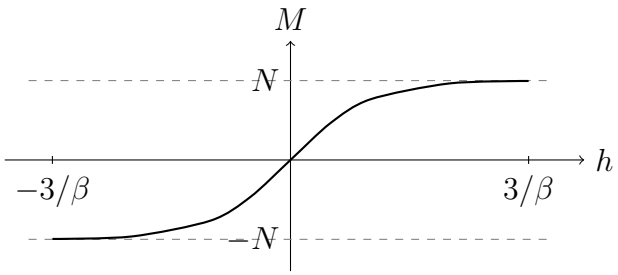
Using the result from (2),

$$\begin{aligned} M &= Nk_B T\frac{\partial}{\partial h} \ln[2\cosh(\beta h)] \\ &= Nk_B T\beta\tanh(\beta h). \end{aligned}$$

Since $\beta = 1/(k_B T)$, we obtain

$$\boxed{M = N \tanh(\beta h)}.$$

The magnetization curve is shown below.



(5)

For $h \ll -1/\beta$, we have

$$M \to -N,$$

which means that almost all spins point down.

For $h = 0$, we have

$$M = 0,$$

which means that spin-up and spin-down states are equally probable, corresponding to maximum disorder in spin orientation.

For $h \gg 1/\beta$, we have

$$M \to N,$$

which means that almost all spins point up.

II

(6)

The Hamiltonian of the system is

$$H = -J \sum_{\langle i,j \rangle} S_i S_j - h \sum_{i=1}^{N} S_i.$$

Let

$$S_i = m + \delta S_i, \qquad S_j = m + \delta S_j.$$

Neglecting the second-order term $\delta S_i \delta S_j$, we obtain

$$\begin{aligned} S_i S_j &= (m + \delta S_i)(m + \delta S_j) \\ &\approx m^2 + m(\delta S_i + \delta S_j) \\ &= m(S_i + S_j) - m^2. \end{aligned}$$

Substituting this approximation into the Hamiltonian, we obtain

$$\begin{aligned} H_{\mathrm{MF}} = & -Jm \sum_{\langle i,j \rangle} (S_i + S_j) \\ & + Jm^2 \sum_{\langle i,j \rangle} 1 - h \sum_{i=1}^{N} S_i. \end{aligned}$$

Since each site has z nearest neighbors, the total number of nearest-neighbor pairs is $Nz/2$.

Furthermore, each spin is counted z times in the first sum, so

$$\sum_{\langle i,j \rangle} (S_i + S_j) = z \sum_{i=1}^{N} S_i.$$

Therefore, we obtain

$$\boxed{H_{\mathrm{MF}} = -Jzm \sum_{i=1}^{N} S_i - h \sum_{i=1}^{N} S_i + \frac{1}{2} NJzm^2}.$$

(7)

The partition function in the mean-field approximation is

$$Z_{\mathrm{MF}} = \sum_{\{S_i\}} e^{-\beta H_{\mathrm{MF}}},$$

where the summation runs over all possible spin configurations.

Substituting the result from (6), we obtain

$$Z_{\mathrm{MF}} = \sum_{\{S_i\}} \exp \left[\beta (Jzm + h) \sum_{i=1}^{N} S_i - \frac{1}{2} \beta NJzm^2 \right].$$

Since the last term is independent of the spin configuration, it can be taken outside the summation:

$$\begin{aligned} Z_{\mathrm{MF}} = & \, e^{-\frac{1}{2}\beta NJzm^2} \\ & \times \sum_{\{S_i\}} \exp \left[\beta (Jzm + h) \sum_{i=1}^{N} S_i \right]. \end{aligned}$$

Notice that the exponential can be factorized into a product over individual spins:

$$\exp \left[\beta (Jzm + h) \sum_{i=1}^{N} S_i \right] = \prod_{i=1}^{N} e^{\beta (Jzm + h) S_i}.$$

Since the spins are independent in the mean-field approximation, we have

$$Z_{\mathrm{MF}} = e^{-\frac{1}{2}\beta NJzm^2} \times \prod_{i=1}^{N} \left(\sum_{S_i=\pm 1} e^{\beta(Jzm+h)S_i} \right).$$

For each spin,

$$\begin{aligned} \sum_{S_i=\pm 1} e^{\beta(Jzm+h)S_i} &= e^{\beta(Jzm+h)} + e^{-\beta(Jzm+h)} \\ &= 2\cosh[\beta(Jzm+h)]. \end{aligned}$$

Therefore, the partition function becomes

$$\boxed{Z_{\mathrm{MF}} = e^{-\frac{1}{2}\beta NJzm^2} [2\cosh\{\beta(Jzm+h)\}]^N}.$$

(8)

The Helmholtz free energy in the mean-field approximation is

$$F_{\mathrm{MF}} = -k_B T \ln Z_{\mathrm{MF}}.$$

Substituting the result from (7), we obtain

$$\begin{aligned} F_{\mathrm{MF}} = & -k_B T \left[-\frac{1}{2}\beta NJzm^2 \right. \\ & \left. + N \ln[2\cosh\{\beta(Jzm+h)\}] \right]. \end{aligned}$$

Since $k_B T = 1/\beta$, we finally get

$$\boxed{F_{\mathrm{MF}} = \frac{1}{2}NJzm^2 - \frac{N}{\beta}\ln[2\cosh\{\beta(Jzm+h)\}]}.$$

(9)

The total magnetization in the mean-field approximation is

$$M_{\mathrm{MF}} = -\left(\frac{\partial F_{\mathrm{MF}}}{\partial h} \right)_{T,m}.$$

Using the result from (8), we obtain

$$M_{\mathrm{MF}} = N\tanh[\beta(Jzm+h)].$$

Since the total magnetization must satisfy

$$M_{\mathrm{MF}} = Nm,$$

we obtain the self-consistent equation

$$\boxed{m = \tanh[\beta(Jzm+h)]}.$$

III

(10)

Now, for $h = 0$, the self-consistent equation is

$$m = \tanh(\beta J z m).$$

Let

$$x = \beta J z m.$$

Then,

$$\frac{x}{\beta J z} = \tanh x.$$

By observing the function $\tanh x$, and noticing that

$$\left.\frac{d}{dx}\tanh x\right|_{x=0} = 1,$$

we can compare the slopes of the two functions at the origin.

If

$$\frac{1}{\beta J z} \geq 1,$$

there is only one solution, $m = 0$.

Otherwise, there are three solutions.

Therefore, the critical temperature satisfies

$$\frac{1}{\beta_c J z} = 1.$$

Using $\beta_c = 1/(k_B T_c)$, we obtain

$$\frac{k_B T_c}{J z} = 1.$$

Thus,

$$\boxed{T_c = \frac{J z}{k_B}}.$$

(11)

Let us rewrite the self-consistent equation using the critical temperature:

$$m = \tanh\left(\frac{T_c}{T} m\right).$$

For temperatures close to T_c, the spontaneous magnetization is small. We can therefore use the approximation

$$\tanh x \approx x - \frac{x^3}{3}.$$

Substituting this approximation, we obtain

$$m \approx \frac{T_c}{T} m - \frac{1}{3} \left(\frac{T_c}{T} \right)^3 m^3.$$

For the nonzero solutions, dividing by m gives

$$1 = \frac{T_c}{T} - \frac{1}{3} \left(\frac{T_c}{T} \right)^3 m^2.$$

Rearranging,

$$\begin{aligned} m^2 &= 3 \left(\frac{T}{T_c} \right)^3 \left(\frac{T_c}{T} - 1 \right) \\ &= 3 \left(\frac{T}{T_c} \right)^2 \frac{T_c - T}{T_c}. \end{aligned}$$

Since T is sufficiently close to T_c, we have

$$\frac{T}{T_c} \approx 1.$$

Therefore, to the lowest order,

$$|m| \approx \sqrt{3} \left(\frac{T_c - T}{T_c} \right)^{1/2}.$$

Thus,

$$\boxed{|m| \propto \left(\frac{T_c - T}{T_c} \right)^{1/2}}.$$